



CODEBOXX

How we built the World's Greatest ChatBot

April 24, 2025

The prestigious RetailTech Breakthrough program in 2025, recognized CodeBoxx as the Builder and operator of the [Best Chatbot Solution of 2025](#). We accepted such recognition with great pride, deep gratitude, and frankly, some surprise. It wasn't immediately clear to our humble company and team how our "Business-First" approach to Agentic Artificial Intelligence (Agentic AI) was pushing boundaries and redefining expectations. When we discovered how our methodology and its results, both in terms of conversion and perceived satisfaction, resonated and stood in the marketplace, we realized the impact we were making for the company using our ChatBot, its brand and its customers. So we decided to fully document, expose and explain our recipe to the broader community: A blueprint, if you will, for building and operating ChatBots the right way; Agents always available, secure at all times, purpose-driven, relevant, capable, precise and constantly fine-tuned to deliver the best user experience-fit and set to evolve and improve as the technology matures.

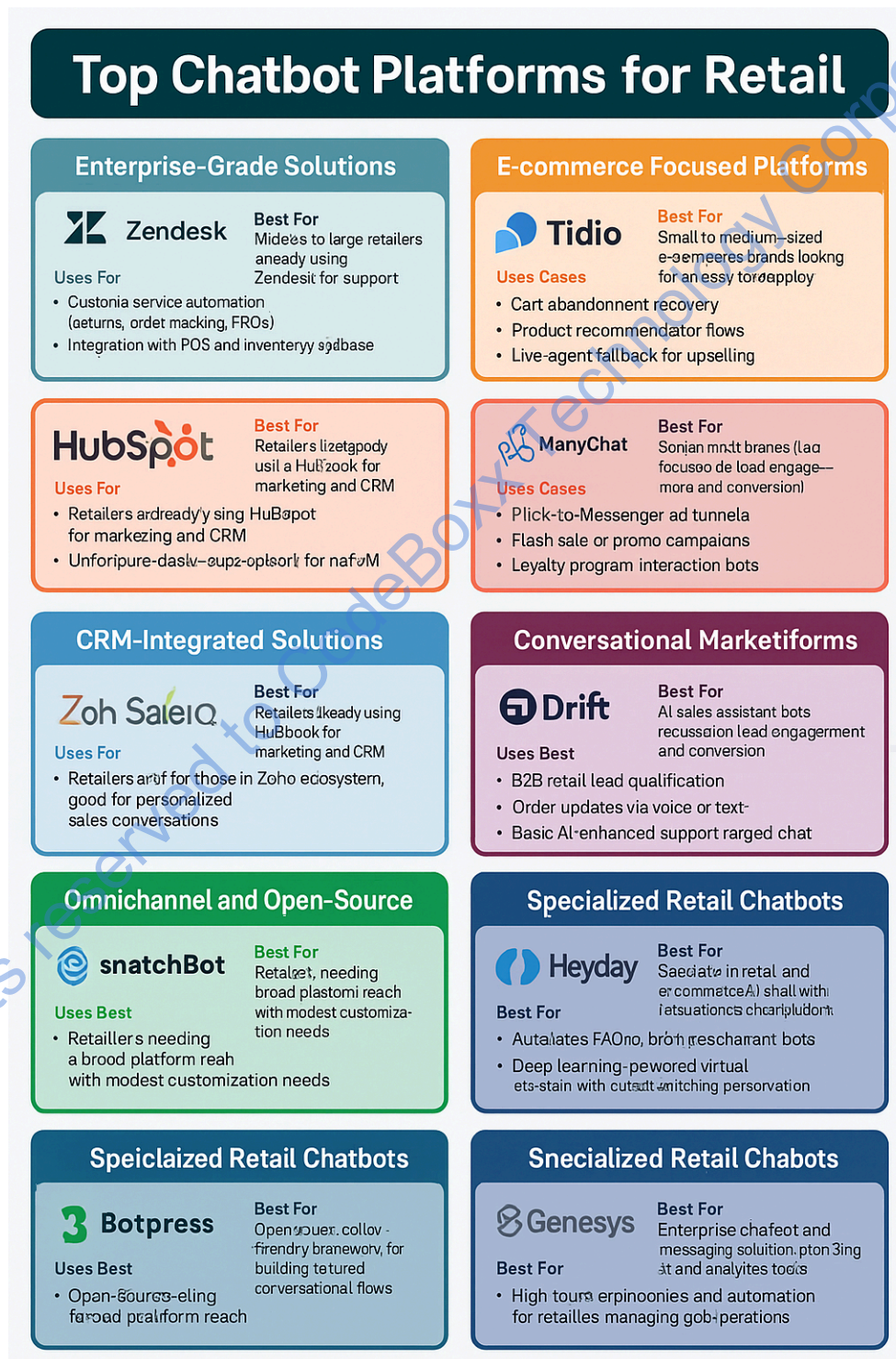


The mission of the annual RetailTech Breakthrough Awards program is to spotlight and celebrate the global innovators who are transforming the retail technology landscape. The RetailTech Breakthrough Awards aim to conduct the industry's most comprehensive analysis of the breakthrough technologies driving innovation and shaping the future of retail. This year's program received thousands of nominations from more than 14 countries worldwide, reflecting the global momentum and impact of retail technology advancements.

CONSUMER BEHAVIOR AND EXISTING SOLUTIONS LANDSCAPE

Most AI agents sound the same because they are trained the same. But when you build around your brand, voice, tone, and edge, you create something that only your business can deploy.

Choosing the best chatbot technology for Retail depends on your specific needs and priorities, but there is already an extensive selection of highly-regarded platforms and some of them have clear strengths in a retail context:



Also relevant in that sector:

LivePerson: AI-powered conversational commerce, real-time messaging across SMS, WhatsApp, web, and apps. Excellent for customer intent detection and personalized product recommendations.

Intercom: Combines bots, live chat, and customer data for proactive support. Highly effective for onboarding, retention, and support with contextual conversations.

Botpress: Open-source, developer-friendly framework for building tailored conversational flows. Ideal for custom logic and integration-heavy retail environments.

Kore.ai: Enterprise-grade NLP engine, scalable across customer service and back-office functions. Supports both text and voice bots for omnichannel retail automation.

Conversica: AI sales assistant bots focused on lead engagement and conversion. Especially valuable for follow-up in high-ticket retail or B2B scenarios.

Zoho SalesIQ: Chatbot + visitor tracking + CRM integration. Seamless fit for those in the Zoho ecosystem; good for personalized sales conversations.

Freshchat (by Freshworks): Unified inbox, AI-based bots, proactive messaging. Cost-effective and scalable for mid-sized retailers aiming to reduce support costs.

Heyday (by Hootsuite): Specializes in retail and e-commerce AI chat, with integrations to Shopify and Magento. Automates FAQs, product discovery, and cart recovery with a retail-first mindset.

Rulai: Deep learning-powered virtual assistant with context-switching and personalization. Suitable for large retailers with complex customer service flows.

Genesys DX (formerly Bold360): Enterprise chatbot and messaging solution, strong AI and analytics tools. High-touch experience and automation for retailers managing global operations.

For over a decade, chatbot platforms like Zendesk, Drift, Ada, and ManyChat effectively served a wide spectrum of retail and service needs by automating narrow, structured interactions. They were rule-based, intent-driven, and modular — fit for the time, scalable within limits, and sufficient for scripted customer support and lead capture flows.

But Generative AI and Large Language Models (LLMs) have irreversibly changed the equation.

These traditional tools, once viewed as cutting-edge, are now increasingly obsolete — not because they ceased functioning, but **because their foundational architecture cannot compete with the adaptability, fluency, and intelligence of modern generative agents.**

WHY OUT-OF-THE-BOX CHATBOTS CAN NO LONGER WIN IN THE LONG RUN

In a market where **customer patience is measured in seconds** and loyalty is shaped by precision and relevance, generic bots will fail to pass the test of today's hyper-demanding digital users. "Out-of-the-box" or licensed chatbot platforms may offer convenience, speed in terms of time-to-market and even maybe some kind of

cost-effectiveness for technology-enabled companies who cannot afford or find the required narrow Data and AI competencies in-house, but all these solutions, no matter how mature, are inherently limited in their ability to scale, adapt, and resonate with user experience needs, aspirations and increasing standards. These general-purpose bots are often:

- **Forcibly Generic:** Built to cover many use cases poorly rather than a few exceptionally well.
- **Rigid:** Heavily template-based, lacking flexibility to reflect brand tone, fast-changing user aspiration, business logic, or adopt industry-specific jargon.
- **Shallow in context:** Unable to fully leverage proprietary business data or adapt to the nuances of customer or internal user interactions that go beyond scripted paths.
- **Vulnerable:** More susceptible to security gaps, especially when users integrate third-party features without deeply understanding data flows and access layers.

This unforgiving user skepticism or its reaction to past traumatising conversations is more and more expressed through what is now commonly called the “Operator syndrome” where common behavior these days consists in for traumatised, intolerant or unforgiving users to just type “Operator” or “I want to speak to a human” right away in their prompts to abort their conversation with an agent powered by Artificial Intelligence. This rejection of the technology is due to the low levels of satisfaction and outcomes provided to the users. Before today, previous generations of chatbots.

The chatbot tools previously listed and described each had their time, but the age of conversational templates and flow builders is ending. What lies ahead is a future led by tailor-made, generative AI agents — smart, connected, context-aware, and fully capable of delivering experiences no rigid chatbot ever could.

The baton has passed. The future is not “AI-powered chatbots” — it’s AI-native assistants.

WHY PURPOSE-BUILT AGENTS DELIVER THE ULTIMATE EXPERIENCE

Custom or Tailor-made AI agents are built around your data, goals, users, and workflows — not the other way around. This makes them vastly superior in:

1. Relevance and Precision

Custom bots ingest domain-specific data, product catalogs, FAQs, transactional logic, and even tone-of-voice guidelines. The result: responses that are **context-aware, brand-aligned, and deeply relevant** — far beyond what an off-the-shelf bot could ever deliver.

2. Security by Design

When you build your own agent:

- You control the architecture, including API access, database queries, and user permissions.
- You can enforce zero-trust policies, rigorous testing protocols, and granular user tracking.
- Sensitive data never flows through unknown third-party servers — no vendor lock-in, no black boxes.

3. Experience Fit

User expectations differ by sector (retail vs. banking vs. healthcare), by geography, and even by device. Custom agents can be tuned to:

- Support multiple languages or dialects
- Provide accessibility enhancements
- Integrate seamlessly into your app or platform UX
- Reflect seasonal, regional, or campaign-specific goals
- Deliver the conversational experience across all channels appropriate to the business while quality and efficiency remain consistent across voice, chat, sms and email

4. Continuous Learning & Adaptation

With full ownership, your AI agent can evolve:

- Ingesting live feedback and real-time analytics
- Tied to customer behavior, sales performance, and support tickets
- Delivering an ever-improving experience — personalized over time as conversations are analyzed and used to reinforce behavior and guidance over future conversations, the quality is therefore never static and always improving leading to sometimes memorable experiences worth taking as feedback.

For **GoodwillFinds** and the **Goodwill movement**, opting for the Tailor-made GEM Chatbot was the natural way to go so the Goodwill movement can remain in control of:

- The narrative conveyed by the voice of Gem
- The data to feed Gem with in order for it to always be in a position to answer questions and inquiries thoroughly and definitively to the satisfaction of customers
- The arms and legs of Gem to always send it where it is relevant to serve the purpose of the interaction and maintain consistency in the outcomes
- The analysis of the conversations both by identified customer-service business owners and AI itself in order to adjust behavior and reinforce the brief and data

LLMs (like GPT-4, Claude, Grok, LLama-3 and Gemini):

- Interpret ambiguous inputs with natural understanding.
- Dynamically generate responses rather than retrieve scripted ones.
- Learn and evolve from proprietary data and context in real time.
- Bridge systems, pulling insights from APIs, databases, and user context in one seamless flow.

In contrast, the rule-based or hybrid bots of the past:

- Require constant human maintenance.
- Break when customers deviate from expected inputs.
- Rely on rigid flows that don't adapt or personalize beyond basic logic trees.
- Often silo customer data rather than intelligently use it.

Even the most advanced pre-LLM tools — like Ada or Kore.ai — while powerful, still lack the foundational fluidity and emergent intelligence that generative models offer. Their frameworks were never designed for what LLMs now make trivial:

- Real-time learning
- Language nuance
- Knowledge retrieval
- Context awareness

CONVERSATIONAL AGENT AND CHATBOTS - THE CODEBOXX BLUEPRINT

In order to build a ChatBot or a conversational agent worthy passing the adoption test of users and the delivery test of business stakeholders, you need to commit to a “Business-First approach”. Adopting a Business-First approach to building an AI conversational agent means rejecting the default impulse to “start with the tech” and instead anchoring everything — the design, architecture, data strategy, and deployment — around your business objectives, pain points, and competitive advantages.

Your Bot Isn't a Chatbot. It's an employee.

A Business-First approach treats your conversational agent like a recruit: you don't train them on everything; you train them on what they need to know to drive outcomes in your business.

- In retail: product discovery, checkout, conversion, upsell, returns automation.
- In healthcare: appointment triage, symptom checking, claims assistance.
- In logistics: dispatch updates, ETA resolution, B2B coordination.

The agent doesn't just talk. It works.

Business Goals Become the AI's Operating System

Instead of retrofitting a generic AI into workflows, Business-First design begins by asking:

“What are the exact tasks or frictions this agent should improve, eliminate, or accelerate?”

From this comes everything else:

- Data ingestion plans tied to business logic.
- APIs wired to real workflows (e.g., refunding orders, modifying bookings).
- Success metrics that matter: ROI, ticket deflection, lead quality, LTV — not just “chat engagement.”

You're not chasing AI novelty. You're engineering business impact

Creativity Where It Counts: Your Unique Advantage

Most AI agents sound the same because they are trained the same. But when you **build around your brand, voice, tone, and edge**, you create something that **only your business can deploy**:

- An AI that sells like your top human rep.
- An assistant that reflects your hospitality values.
- A support agent that understands your loyalty tiers, offers, and SKUs.

This isn't plug-and-play. It's craft. Business-first means creative expression *and* operational precision.

The Right Problems Get Solved — Not Just the Noisy Ones

Tech-first teams often build for:

- Whatever's easiest to automate.
- Whatever data is cleanest.
- Whatever tool demos best.

Business-first teams build for:

- **What's killing margins**
- **What drives customer churn**
- **What competitors can't copy**

In other words: **high-leverage problems**, not low-hanging fruit. Like everywhere else in your business, you win when you agree and commit to do the hard things.

A Business-First approach is what will give a soul to your Conversational Agent.

It produces an AI agent that:

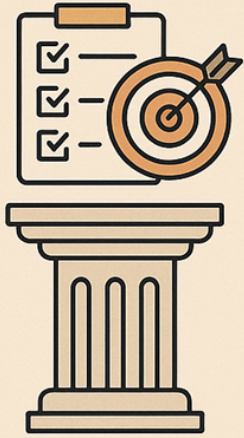
- Justifies its cost by delivering value
- Knows your customer like your team does
- Becomes part of your strategic fabric, not just a novelty layer

And in a world where every company will soon have access to the same foundational models, **your advantage won't be the model — It will stem from your mastery of the following Four Pillars of Award-Winning conversational Agent**

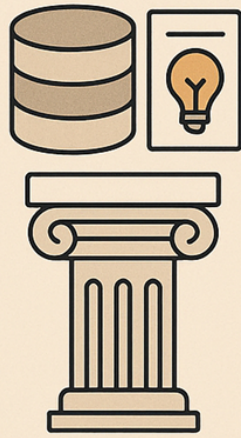
CONVERSATIONAL AGENT AND CHATBOTS - THE FOUR PILLARS TO MASTER

Behind every truly exceptional AI agent, one that feels intelligent, responsive, and indispensable, lies deliberate design. Not just lines of code, but intention, discipline, and the mastery of four essential pillars. Each pillar supports a domain of expertise that, when perfected, elevates a generative AI model from good to groundbreaking. This white paper series explores these pillars in depth, offering practitioners, strategists, and technologists a framework for excellence in AI conversation design.

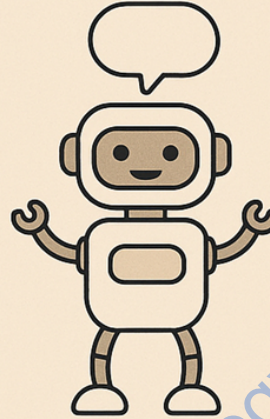
FOUR PILLARS TO MASTER FOR AN AWARD-WINNING GENERATIVE AI CONVERSATION MODEL



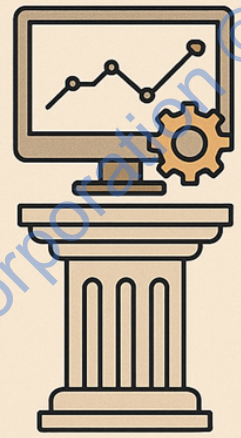
1. BRIEF PERFECTLY ON PURPOSE



2. FEED IT WITH DATA AND KNOWLEDGE



3. GIVE IT ARMS AND LEGS



4. MONITOR AND ADJUST BEHAVIOR CONSTANTLY THROUGH AUTOMATION

“Don’t even get started if you are not committed to give each pillar the level of caring it deserves or if you are not clear about your definition of success”

1. Brief Perfectly on Purpose

The soul of a great AI Agent begins with clarity in purpose. This white paper explores how purpose-driven prompts, constraints, and role definitions serve as the blueprint for reliable AI behavior. We will demonstrate how briefing is not just a setup but a strategy. We will also spend time explaining how to make sense of all the mature large language models out there and propose an approach allowing to always leverage the ones servicing the agent’s purpose in the best way possible.

2. Feed it with Data and Knowledge

AI learns and remembers what it is fed. But feeding it isn't just about volume—it's about precision, relevance, and curated insight. In this white paper, we dive into data architecture, retrieval-augmented generation (RAG), and how to inject structured and unstructured knowledge that gives the model its mind.

3. Give it Arms and Legs

Generative AI shouldn't just talk—it should act. This white paper unveils how to integrate tools, APIs, workflows, and triggers so your AI can make decisions, complete tasks, and deliver real-world value. This is how language meets impact to satisfy aspirations. An agent needs the legs to go wherever needed to gather context-specific information and it uses its arms to take action based on de-risked decisions.

4. Monitor and Adjust Behavior Constantly through Automation

AI must meet expectations and fulfill aspirations to be adopted and remain useful over time. The final white paper in the series shows how to implement a complete operational framework beyond the buildout to measure success, monitor interactions and qualify outcomes. Through a tight web of real-time feedback loops, behavioral telemetry, reinforcement guidance and self-healing prompts, your agent adapts automatically to new use cases, isolates exceptions, edge cases, and even highlights changing expectations.

Final Remarks

If you've reached this point in your reading, it means you have a genuine thirst and a real hunger for building and operating an effective custom AI chatbot with the technology, tools and capabilities available to you today. A next-generation Chatbot that is tailored around your business's unique data, goals, users and workflows. You've likely absorbed new ideas, challenged existing assumptions, and perhaps even imagined what a tailored AI agent could do for your own business.

That's exactly what we hoped for.

However, we can't overstate the importance of adhering to and respecting the *Business-First delivery model* in its entirety. It has been our North Star for every endeavour we undertake at CodeBoxx. This approach treats your AI agent, from conception to deployment, akin to a valuable employee or to some extent, as a key member of your personnel.

This 24/7 relentless employee working in perpetuity, has the focus, propensity and power to solve targeted business pain points, which in turn drive both top- and bottom-line growth, as well as, reduce customer churn. The Gem chatbot we conceived and deployed at GoodwillFinds is a testament to this approach. Up to this day, it is able to successfully solve and handle 78% of all customer inquiries autonomously. The remaining cases are escalated to the CX team for direct engagement and high-value interactions with human agents. And like any great employee, Gem gets smarter and more intuitive with time, learning, adapting, and continuously improving in real-time through real customer interactions.

The promise of generative AI agents transforming how brands serve, sell, and support is immense, but so are the risks of getting it wrong and so, we would like to offer a word of realism:

When you order the Four Pillars, you are in fact, ordering to receive our secret sauce for transforming an idea into an operating AI solution. The Four Pillars dives deeper and further but we also recognize that absorbing extensive material can be time-consuming and, while reading the full dissertation is enriching, knowledge alone isn't enough and won't build an agentic AI that delivers results. Just like with fitness, we all "know" what to do: eat clean, hit the gym, put in the reps. We read about it, talk about it, even plan for it. But how many of us who crave that lean, fit, enviable result actually show up and *do* the work?

The space we are operating in is not trivial. It still takes experienced doers, those people who are not to be replaced by AI anytime soon, teams who've made the mistakes, learned the hard lessons, bounced against the limitations and already shipped solutions that outperform. CodeBoxx technologists have walked this path - a multitude of times. And we offer to walk this rigorous path with you, just as a fitness coach would. We consider ourselves as premier tech partners, advisors and builders with a pedigree grounded on over 60+ years of cumulated tech leadership expertise and know-how.

The truth is, you could in theory build an agentic AI all by yourself, that's why we are sharing the recipe with you and you are certainly able to pull it off and we genuinely respect those who try. But as a company who's navigated the steep learning curve and has been through endless iterations and found what works and what doesn't, we would like to let you know that we are also here and glad to assist. We want to help you build an AI that will command the kind of impact and results your business deserves.

So, whether you are part of the 25% of companies who are expected to launch agentic AI pilots this year or the 50% who expect to do so by 2027, as written in Deloitte *2025 Global TMT Predictions*, or whether you're a startup aiming to build your first AI-native experience, or an enterprise looking to evolve your existing conversational stack into something worthy of your brand's future, please know this: You are not condemned to do it alone if you don't want to or don't have the entire required skillset.

This is our way of sharing the hard-earned DIY knowledge we have accumulated over the past months with those who are ready to take the next step in bringing AI agents to life with a Business-First mindset, integrity and put a voice out there worth talking to.

Sincerely,

Nicolas Genest and the CodeBoxx Team

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